RAFFLES INSTITUTION
Year 6 TIMED PRACTICE 2025
Higher 2

CANDIDATE NAME						CLASS INDEX NUMBER	
CLASS	2	5	S	0			

PHYSICS **9749**

Section A **1 July 2025**

1 hour

You must answer on the multiple choice answer sheet

You will need: Multiple choice answer paper
Soft clean eraser
Soft pencil (type 2B is recommended)

INSTRUCTIONS

- There are **thirty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in **soft pencil** on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
- Write in soft pencil.
- Write your name, class and index number on the multiple choice answer sheet in the spaces provided.
- Do **not** use correction fluid or tape.
- You may use an approved calculator.

INFORMATION

- The total mark for this section is 30.
- Each correct answer will score one mark.
- Any rough working should be done on this question paper.

Data

speed of light in free space	c	=	3.00×10^{8} m s^{-1}
permeability of free space	μ_0	=	$4\pi \times 10^{-7}$ H m^{-1}
permittivity of free space	ε_0	=	8.85×10^{-12} F m^{-1}
		=	$(1/(36\pi)) \times 10^{-9}$ F m^{-1}
elementary charge	e	=	1.60×10^{-19} C
the Planck constant	h	=	6.63×10^{-34} J s
unified atomic mass constant	u	=	1.66×10^{-27} kg
rest mass of electron	m_e	=	9.11×10^{-31} kg
rest mass of proton	m_p	=	1.67×10^{-27} kg
molar gas constant	R	=	8.31 J K^{-1} mol^{-1}
the Avogadro constant	N_A	=	6.02×10^{23} mol^{-1}
the Boltzmann constant	k	=	1.38×10^{-23} J K^{-1}
gravitational constant	G	=	6.67×10^{-11} N m^{2} kg^{-2}
acceleration of free fall	g	=	9.81 m s^{-2}

Formulae

uniformly accelerated motion	s	=	$ut + \frac{1}{2}at^2$
	v^2	=	$u^2 + 2as$
work done on/by a gas	W	=	$p\Delta V$
hydrostatic pressure	p	=	ρgh
gravitational potential	ϕ	=	$-Gm/r$
temperature	T / K	=	T / °C + 273.15
pressure of an ideal gas	p	=	$\frac{1}{3}\frac{Nm}{V}\langle c^2 \rangle$
mean translational kinetic energy of an ideal gas molecule	E	=	$\frac{3}{2}kT$
displacement of particle in s.h.m.	x	=	$x_0 \sin \omega t$
velocity of particle in s.h.m.	v	=	$v_0 \cos \omega t = \pm\omega\sqrt{x_0^2 - x^2}$
electric current	I	=	$Anvq$
resistors in series	R	=	$R_1 + R_2 + \ldots$
resistors in parallel	$1/R$	=	$1/R_1 + 1/R_2 + \ldots$
electric potential	V	=	$\frac{Q}{4\pi\varepsilon_0 r}$
alternating current/voltage	x	=	$x_0 \sin \omega t$
magnetic flux density due to a long straight wire	B	=	$\frac{\mu_0 I}{2\pi d}$
magnetic flux density due to a flat circular coil	B	=	$\frac{\mu_0 NI}{2r}$
magnetic flux density due to a long solenoid	B	=	$\mu_0 nI$
radioactive decay	x	=	$x_0 \exp(-\lambda t)$
decay constant	λ	=	$\ln 2 / t_{1/2}$

Section A

1 A particle is oscillating horizontally with simple harmonic motion.

What is ratio of the kinetic energy to the potential energy of the oscillations when the particle is midway between the equilibrium position and amplitude position?

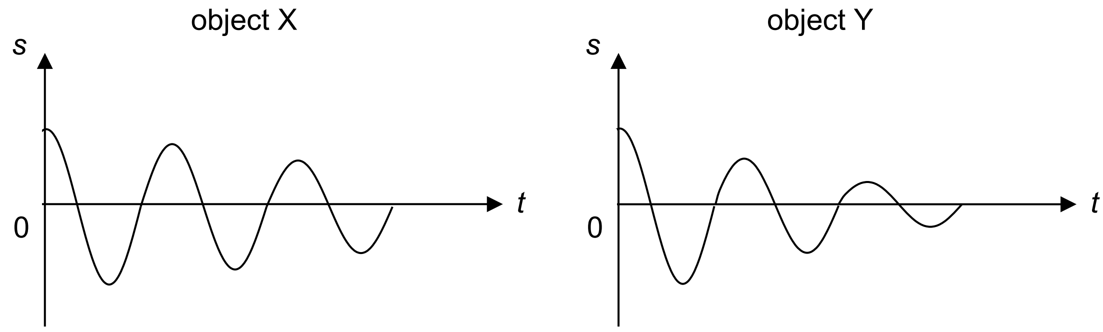
A 1:3 **B** 1:1 **C** 3:1 **D** 1:7

2 The suspension system of a fully loaded car has a natural frequency of 1.20 Hz. When the car is driven over a series of equally spaced bumps at a speed of 5.0 m s^{-1}, the amplitude of vibration becomes much larger.

What is the separation of the bumps?

A 1.1 m **B** 2.3 m **C** 4.2 m **D** 6.0 m

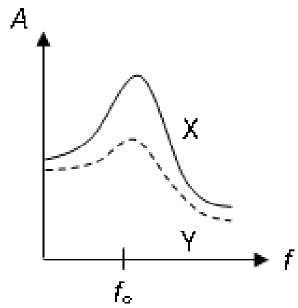
3 Two objects X and Y of equal mass and natural frequency f_0 are given the same initial displacement and are then released. The graphs show the variation with time t of their displacements s.



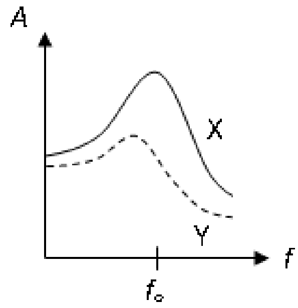
X and Y are then subjected to driving forces of the same constant amplitude and of variable frequency f.

Which of the following sets of graphs represents the variation with f of the amplitudes A of X and Y?

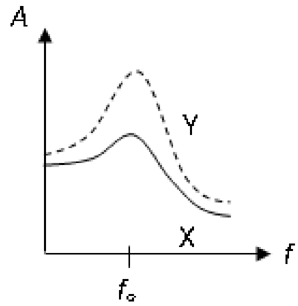
A



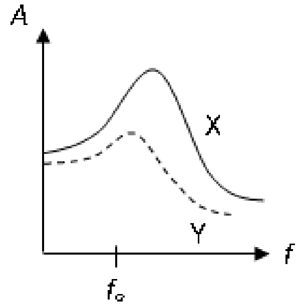
B



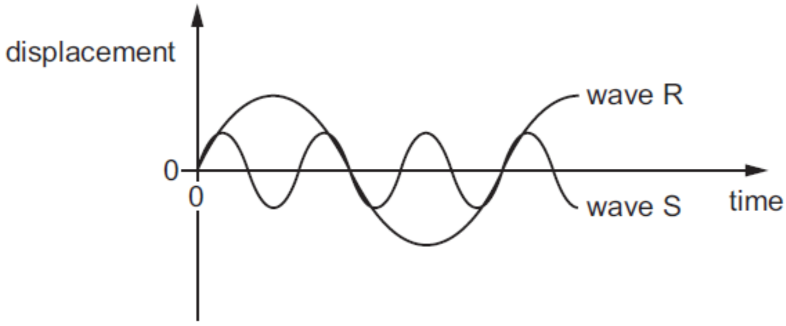
C



D



4 The diagram shows two waves R and S.

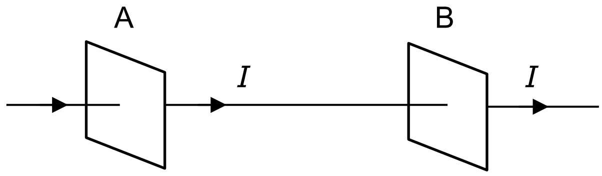


Wave R has an amplitude of 8 cm and a period of 30 ms.

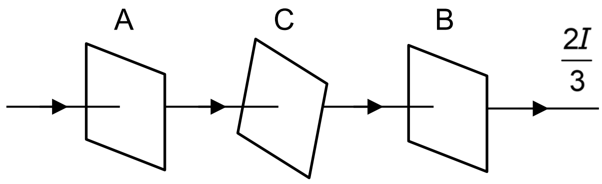
What are the amplitude and the period of wave S?

	amplitude / cm	period / ms
A	2	10
B	2	90
C	4	10
D	4	90

5 When a narrow beam of plane-polarised light passes through an ideal polarising filter A, the light intensity is observed to be I. When an identical polarising filter B is placed behind A, it is observed that the light intensity beyond B is still I.



A third identical polarising filter C is now inserted between A and B, causing the light intensity emerging from B to become $\frac{2I}{3}$.



What is the angle between the axes of polarisation of polarising filters A and C?

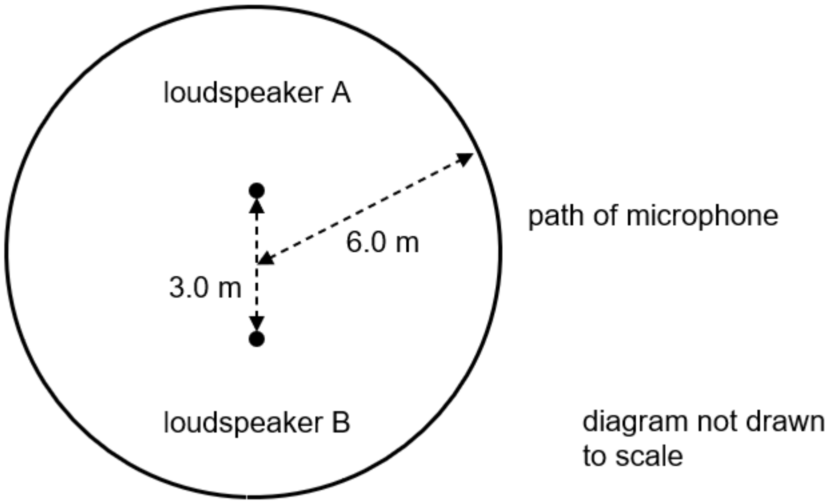
A 25.4° **B** 35.3° **C** 48.2° **D** 55.6°

6 Two sources of radio waves are at a distance of 1.0×10^{15} m from Earth. The sources are separated by a distance of 1.0×10^{12} m and emit radio waves of wavelength 0.030 m.

What is an estimate of the diameter of a dish of a radio telescope on Earth that can just resolve the two sources?

A 3.0×10^{-5} m **B** 3.0×10^{-2} m **C** 3.0 m **D** 30 m

7 Two small identical loudspeakers are placed 3.0 m apart. The loudspeakers emit sound waves that are in phase and of wavelength 1.0 m uniformly in all directions.



A microphone is moved in a circular path of radius 6.0 m. The centre of the circular path coincides with the midpoint of the two loudspeakers.

How many maxima are detected by the microphone in one complete round?

A 8 **B** 10 **C** 12 **D** 14

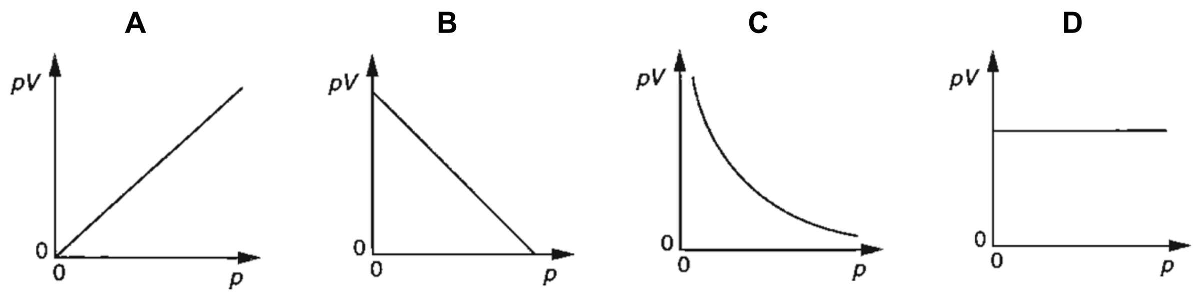
8 Monochromatic light of wavelength 5.30×10^{-7} m is incident normally on a diffraction grating. The first order maximum is observed at an angle of 15.4° to the direction of the incident light.

What is the angle between the first and second orders diffraction maxima?

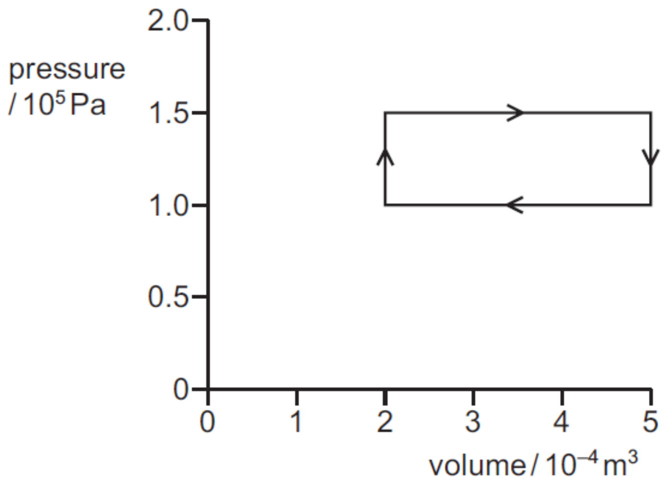
A 7.7° **B** 15.4° **C** 16.7° **D** 32.1°

9 In an experiment to investigate the relationship between the volume V of a fixed mass of an ideal gas and its pressure p, a graph of pV against p is plotted.

Which graph shows the correct relationship at constant temperature?



10 A fixed amount of a gas undergoes a series of changes to its pressure and volume.



What is the total work that is done on the gas in one cycle?

A –45 J **B** –15 J **C** 15 J **D** 30 J

11 The first law of thermodynamics, when applied to a system, can be represented by the equation:

$$\Delta U = q + w$$

where ΔU is the increase in internal energy of the system,
q is the heat supplied to the system and
w is the work done on the system.

A fixed mass of ideal gas at high pressure is contained in a balloon. The balloon suddenly bursts, and the gas expands and cools.

Which row describes the values of ΔU, q and w in this situation?

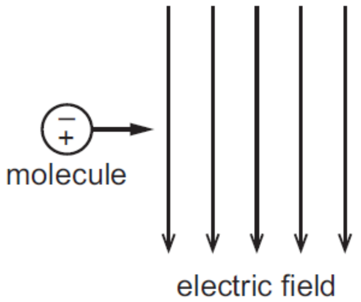
	ΔU	q	w
A	negative	negative	positive
B	negative	zero	negative
C	positive	zero	negative
D	positive	negative	positive

12 The charge of the uranium nucleus is 1.5×10^{-17} C and the charge of the alpha particle is 3.2×10^{-19} C.

What is the electric force between a uranium nucleus and an alpha particle when separated by a distance of 1.0×10^{-13} m?

A 4.3×10^{-17} N **B** 4.3×10^{-13} N **C** 4.3 N **D** 4.3×10^{10} N

13 A molecule behaves as an electric dipole consisting of two equal point charges of opposite sign, separated by a fixed distance. The molecule moves with constant horizontal velocity as it enters a vertical uniform electric field.



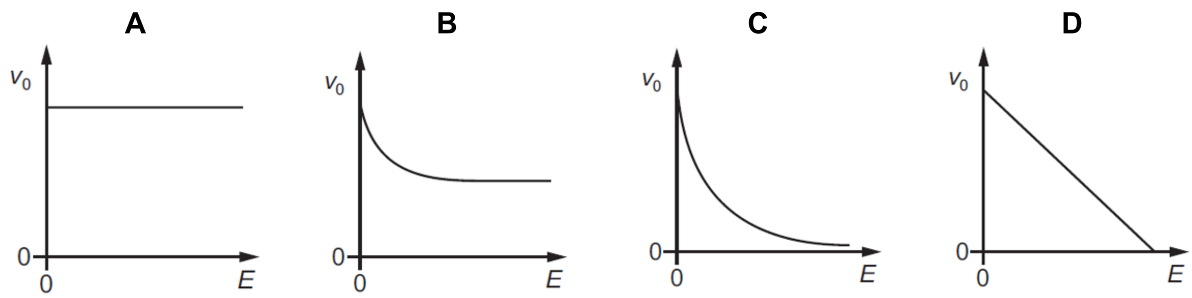
The positive and negative charges of the molecule enter the field at the same time.

Which of the following describes the velocity of the molecule in the field?

	horizontal component of velocity	vertical component of velocity
A	constant	increases
B	constant	zero
C	increases	increases
D	increases	zero

14 A positively charged oil droplet falls in air in a uniform electric field that is vertically upwards. The droplet has a terminal speed v_0 and the electric field strength is E. The magnitude of the force due to air resistance acting on the droplet is proportional to the speed of the droplet.

Which graph shows the variation with E of v_0?



15 The drift velocity of the electrons in a cylindrical metal conductor is v when the current in the conductor is I.

What is the new drift velocity of the electrons in another cylindrical conductor made of the same material with twice the diameter and twice the current?

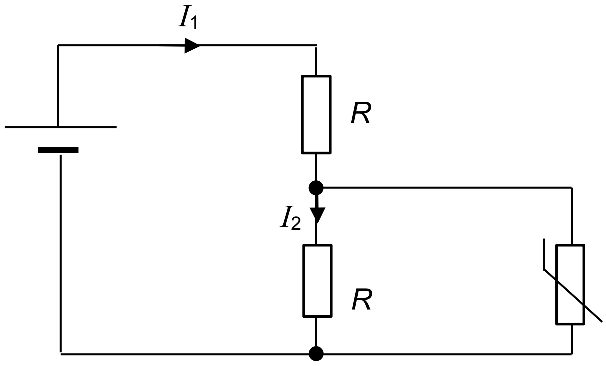
A $\dfrac{v}{2}$ **B** v **C** $2v$ **D** $4v$

16 Bulb A, with a rating of 20 W, 120 V has a filament wire that is of the same length as the filament wire in bulb B, rated 20 W, 6 V. The filament wires of both the bulbs are made from tungsten.

What is the value of the ratio $\dfrac{\text{diameter of filament in bulb A}}{\text{diameter of filament in bulb B}}$?

A 2.5×10^{-3} **B** 5.0×10^{-2} **C** 20 **D** 400

17 At room temperature, the resistance of a negative temperature coefficient thermistor is *R*. It is connected to a circuit with two resistors, each of resistance *R*. The currents in the two resistors are I_1 and I_2 as shown.

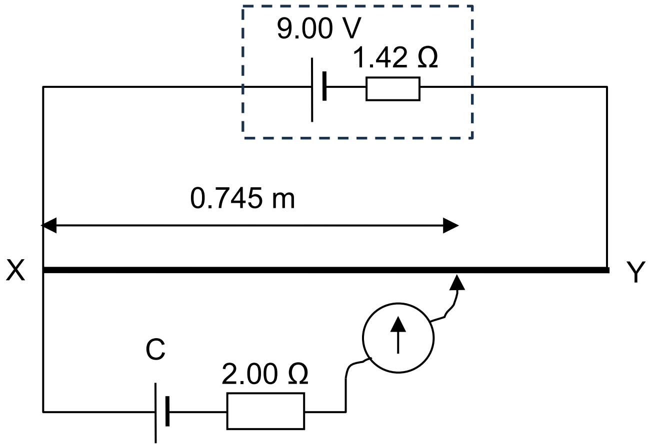


How do the currents change when the temperature of the thermistor increases?

	I_1	I_2
A	increases	increases
B	increases	decreases
C	decreases	increases
D	decreases	decreases

18 A power supply of e.m.f. of 9.00 V and internal resistance of 1.42 Ω is connected across a uniform resistance wire XY. The wire XY has length 1.00 m and resistance 8.30 Ω.

A cell C of negligible internal resistance, a resistor of 2.00 Ω and a galvanometer are connected to wire XY. A balance length of 0.745 m is obtained.



What is the e.m.f. of cell C?

A 1.96 V **B** 5.73 V **C** 6.71 V **D** 7.69 V

19 An alternating current I varies with time t according to the equation:

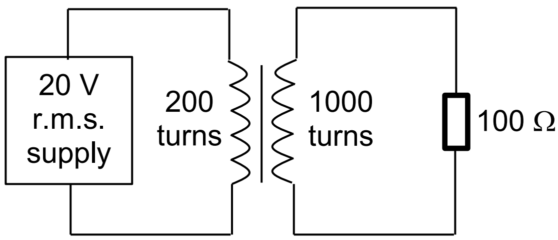
$$I = 5.0\sin\left(\sqrt{3}\,t\right)$$

where I is measured in amperes and t is measured in seconds.

What is the mean power dissipated in a resistive load of 10 Ω?

A 130 W **B** 250 W **C** 380 W **D** 750 W

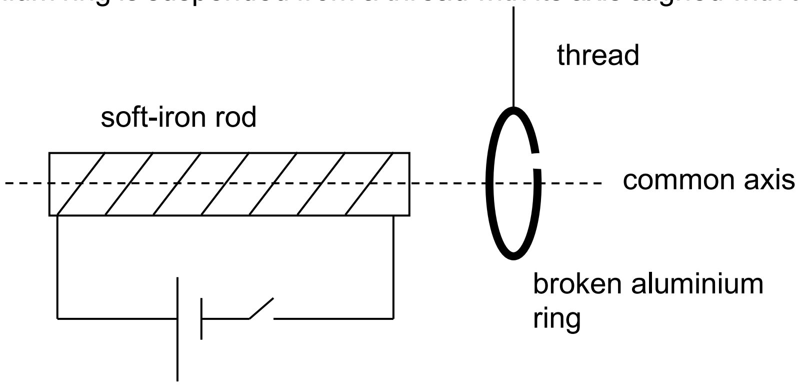
20 The primary coil of an ideal transformer has 200 turns and is connected to a power supply of root-mean-square voltage of 20 V. The secondary coil has 1000 turns and is connected to a resistor of resistance 100 Ω as shown in the diagram.



What is the peak current in the primary coil?

A 1.0 A **B** 1.4 A **C** 5.0 A **D** 7.1 A

21 A solenoid is connected to a cell and a switch. A soft-iron rod is inserted into the solenoid. A broken aluminium ring is suspended from a thread with its axis aligned with that of the solenoid.



What will happen to the aluminium ring when the switch is closed?

A It will be repelled by the solenoid.

B It will be attracted to the solenoid.

C It will be repelled by the solenoid before being attracted to it.

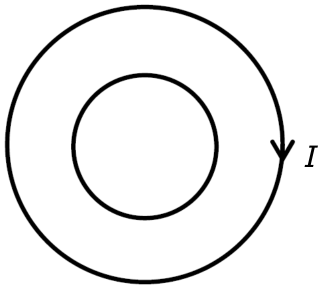
D It will remain at rest.

22 An ion of charge $2q$ moves in a uniform magnetic field of flux density B in a circle of radius r at a speed v.

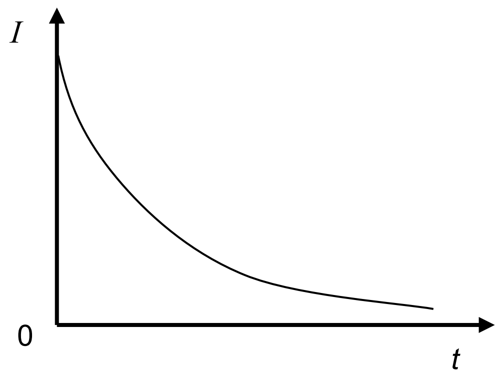
What is the flux density of the field required for an ion of charge q of the same mass to move in a circle of half the radius at the same speed?

A $0.25B$ **B** $0.5B$ **C** $2B$ **D** $4B$

23 Two separate circular wire loops are concentric and lie in the same plane as shown.



The current in the outer loop is clockwise and decreasing with time as shown.



Which row describes the current induced in the inner loop?

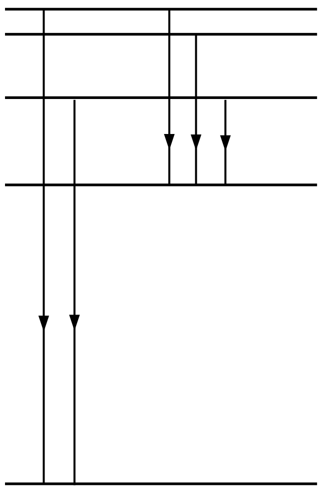
	magnitude of current	direction of current
A	decreasing	clockwise
B	decreasing	anti-clockwise
C	increasing	clockwise
D	increasing	anti-clockwise

24 A circular coil of diameter 7.0 cm and 30 turns is placed so that the plane of the coil is perpendicular to the direction of a uniform magnetic field. The flux density of the field is changed from 20 mT to 60 mT in 0.50 s.

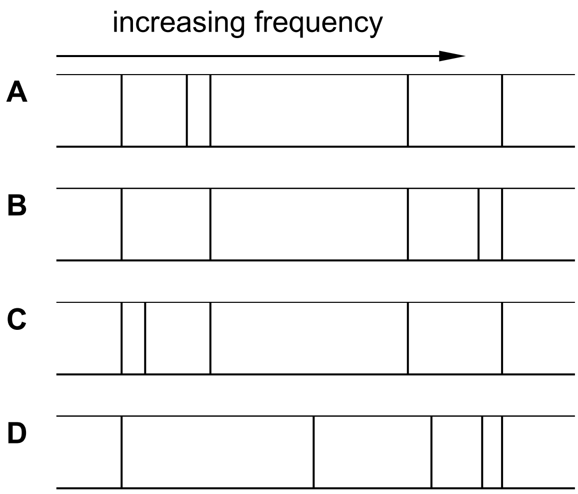
What is the average e.m.f. induced in the coil?

A 0.31 mV **B** 4.6 mV **C** 9.2 mV **D** 37 mV

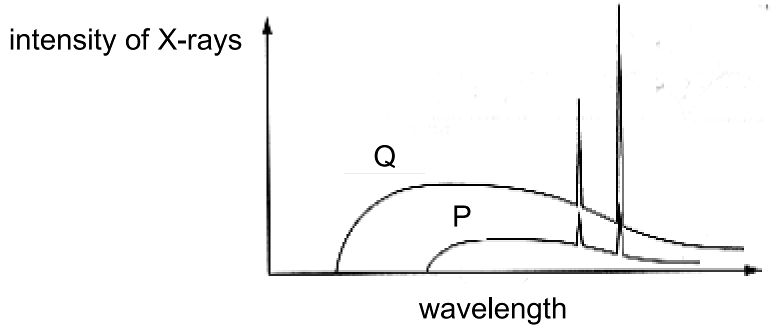
25 The diagram shows five energy levels of an atom. Five possible transitions between the levels are indicated. Each transition produces a photon.



Which spectrum corresponds most closely to the transitions shown?



26 X-ray spectra are taken from two X-ray tubes P and Q. The variation of intensities of the X-rays with wavelength for both X-ray tubes are shown.



What can be deduced from the graph?

A X-ray tube Q has the higher voltage applied to it and the target materials in both tubes are the same.

B X-ray tube Q has the higher voltage applied to it and the target materials in both tubes are different.

C X-ray tube P has the higher voltage applied to it and the target materials in both tubes are the same.

D X-ray tube P has the higher voltage applied to it and the target materials in both tubes are different.

27 A proton has kinetic energy of 1.0 MeV, and its momentum has an uncertainty of 1.0%.

What is the minimum uncertainty in its position?

A 7.0×10^{-13} m **B** 1.1×10^{-12} m **C** 2.9×10^{-12} m **D** 1.1×10^{-11} m

28 Which of the following is not a conclusion drawn from the Rutherford α-particle scattering experiment?

A Most of the atom is empty space.

B The nucleus of the atom is very small and charged.

C Most of the mass of the atom is concentrated in the nucleus.

D Negatively charged electrons revolve around the positively charged nucleus.

29 A slow moving neutron reacts with a stationary nucleus of boron-10 to form a nucleus of lithium-7. An alpha particle and gamma ray are emitted. The reaction is represented by the equation shown.

$$^{10}_{5}\mathrm{B} + {}^{1}_{0}\mathrm{n} \rightarrow {}^{7}_{3}\mathrm{Li} + {}^{4}_{2}\mathrm{He} + \gamma$$

The nuclear binding energies are:

$^{10}_{5}\mathrm{B}$: 64.94 MeV

$^{7}_{3}\mathrm{Li}$: 39.25 MeV

$^{4}_{2}\mathrm{He}$: 28.48 MeV

In one reaction, the total kinetic energy of the lithium-7 nucleus and alpha particle is 2.31 MeV.

What is the energy of the gamma ray emitted?

A 0.480 MeV **B** 2.79 MeV **C** 25.7 MeV **D** 261 MeV

30 In order to detect a leak in a water-pipe buried 0.4 m below the surface of a soccer field, an engineer wishes to add a radioactive isotope to the water.

Which of the following radioactive emitters should be chosen?

	emitter	half-life
A	β	a few hours
B	β	several years
C	α	a few hours
D	α	several years